

SE LAB 3: Architectural Selection Problem

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Architecture Selection

We chose Layered Architecture for the Corporate Mental Wellness & Counseling Platform.

Architectural Choice

We selected the **Layered Architectural Style**, consisting of three main layers: **Presentation Layer, Business Layer, and Data Layer**. The Presentation Layer handles interactions with corporate employees and licensed therapists. The Business Layer manages appointment booking, stress assessment, counseling sessions, payments, and notifications. The Data Layer manages the storage of user, therapist, appointment, assessment, session, and payment information.

Two Reasons

1. Separation of responsibilities:

The platform contains different functions such as anonymous employee access, stress assessment, appointment booking, therapist session management, payment processing, and notifications. Layered Architecture separates these responsibilities into different layers, making the system easier to develop, maintain, and modify.

2. Support for confidential counseling workflows:

The platform must allow employees to access counseling services without revealing their corporate identity to therapists. Separating the presentation, business logic, and data layers helps keep confidential employee information and counseling operations isolated and properly managed.

Security Advantage

Layered Architecture improves security by preventing users from directly accessing the database. Employees and therapists interact with the system through the Presentation and Business layers, where authentication, authorization, validation, and access controls can be applied before sensitive information is accessed. This helps protect anonymous employee profiles, stress assessment results, counseling session details, and payment records.

Performance Benefit

Layered Architecture can improve performance by **centralizing and optimizing database access** through the Data Layer. Business operations can request only the required information instead of allowing direct and unnecessary database access. This helps reduce redundant queries and allows database operations to be optimized independently.